

Uncertainty and Model Checking for Atmospheric Inversions

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2026 BEACO2N Partner Meeting

What we know, and what we want to know

For the inversions we consider, we know:

$\mathbf{y}$: Observed measurements dimension M

$\mathbf{x}$: Unobserved flux dimension N

$\mathbf{H}$: Footprints dimension $M \times N$

$\mathbf{B}$: Prior covariance dimension $N \times N$

$\mathbf{R}$: Observation error dimension $M \times M$

$\mathbf{m}$: Prior inventory dimension N

$\mathbf{x} \sim \mathcal{N}(\mathbf{m}, \mathbf{B})$ Actual fluxes are smooth, centered at inventory

$\mathbf{y}|\mathbf{x} \sim \mathcal{N}(\mathbf{H}\mathbf{x}, \mathbf{R})$ The footprints transport $\mathbf{y}$, then observe with some error

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Would also like, without a ton of extra computation:

- Measurements of posterior uncertainty. (E.g., how large is $\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x})$?)
- Measurements of prior sensitivity (E.g., what if the length scale of $\mathbf{B}$ changed?)
- Measurements of data influence (E.g., how much is my data, and how much my prior?)

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Takeaway: All these are available from stuff you computed to get $\hat{\mathbf{x}}$.

Data generating process:

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$\mathbf{y}|\mathbf{x} \sim \mathcal{N}(\mathbf{H}\mathbf{x}, \mathbf{R})$ The footprints transport $\mathbf{y}$, observed with some error

Inference: $\mathcal{P}(\mathbf{y}, \mathbf{x})$ is Gaussian $\Rightarrow \mathcal{P}(\mathbf{x}|\mathbf{y})$ is Gaussian.

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \quad \text{where}$$

$$\hat{\mathbf{x}} = \mathbf{m} + (\mathbf{H}\mathbf{B})^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{H}\mathbf{m})$$

$$\hat{\mathbf{V}} = \mathbf{B} - (\mathbf{H}\mathbf{B})^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1} (\mathbf{H}\mathbf{B})$$

Data generating process:

$$\begin{aligned}\mathbf{x} &\sim \mathcal{N}(\mathbf{m}, \mathbf{B}) && \text{Actual fluxes are smooth, centered at inventory} \\ \mathbf{y}|\mathbf{x} &\sim \mathcal{N}(\mathbf{H}\mathbf{x}, \mathbf{R}) && \text{The footprints transport } \mathbf{y}, \text{ observed with some error}\end{aligned}$$

Inference: $\mathcal{P}(\mathbf{y}, \mathbf{x})$ is Gaussian $\Rightarrow \mathcal{P}(\mathbf{x}|\mathbf{y})$ is Gaussian.

$$\begin{aligned}\mathcal{P}(\mathbf{x}|\mathbf{y}) &= \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \quad \text{where} \\ \hat{\mathbf{x}} &= \mathbf{m} + (\mathbf{HB})^\top (\mathbf{HBH}^\top + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{Hm}) \\ \hat{\mathbf{V}} &= \mathbf{B} - (\mathbf{HB})^\top (\mathbf{HBH}^\top + \mathbf{R})^{-1} (\mathbf{HB})\end{aligned}$$

Inversions estimate $\hat{\mathbf{x}} = \mathbb{E}_{\mathcal{P}(\mathbf{x}|\mathbf{y})} [\mathbf{x}]$. This is slow because of

- Forming $\mathbf{H}$ and $\mathbf{B}$ in memory
- Computing $\mathbf{HB}$ and $\mathbf{HBH}^\top$
- Inverting $(\mathbf{HBH}^\top + \mathbf{R})$

Sequoia has sped these up, and also saved to disk what she can.

Assume we can re-use these computations. Let's get posterior uncertainty, prior sensitivity, and data influence.

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\hat{\mathbf{x}}, \hat{\mathbf{V}}) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Problem: Obviously we can't compute $\hat{\mathbf{V}}$, it is huge.

Idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

Posterior uncertainty

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Example: Want s , the total emissions in hour t , summed over all spatial locations.

Let $\mathbf{a}$ be an N -vector with:

$$\mathbf{a}_n = \begin{cases} 1 & \text{If observation } n \text{ was at time } t \\ 0 & \text{otherwise} \end{cases}$$

Then

$$s = \sum_{n=1}^N \mathbf{a}_n \mathbf{x}_n = \mathbf{a}^\top \mathbf{x} \Rightarrow$$

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(s) = \text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}^\top \text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x}) \mathbf{a}$$

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The covariance of the scalar s is a linear combination of stuff we computed for $\hat{\mathbf{x}}$:

$$\text{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(s) = \mathbf{a}^\top \underbrace{\mathbf{B}}_{\text{done}} \mathbf{a} - (\underbrace{\mathbf{HB}}_{\text{done}} \mathbf{a})^\top \underbrace{(\mathbf{HBH}^\top + \mathbf{R})^{-1}}_{\text{done}} (\underbrace{\mathbf{HB}}_{\text{done}} \mathbf{a})$$

Variance (Tentative preliminary results!)

Posterior Variance Results

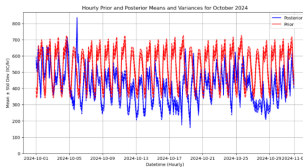


Figure 11: Hourly posterior mean and variance for a month of inversions

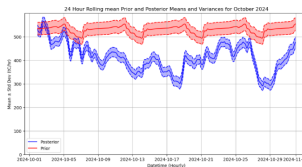


Figure 12: Smoothed hourly posterior mean and variance for a month of inversions

The prior $\mathbf{B}$ depends on quantities like ρ , an assumed lengthscale. So we might write

$$\hat{\mathbf{x}}(\rho) = \mathbf{m} + (\mathbf{H}\mathbf{B}(\rho))^{\top} (\mathbf{H}\mathbf{B}(\rho)\mathbf{H}^{\top} + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{H}\mathbf{m})$$

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- Re-computing $\hat{\mathbf{x}}(\rho)$ for a new lengthscale is expensive (a whole new problem).
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Problem:

- Re-computing $\hat{\mathbf{x}}(\rho)$ for a new lengthscale is expensive (a whole new problem).
- There actually can be lots of hyperparameters, and you don't know which matter.

Idea: Starting from a solution at the initial guess ρ_0 , form a Taylor series expansion

$$\hat{\mathbf{x}}(\rho) \approx \hat{\mathbf{x}} + \frac{\partial \hat{\mathbf{x}}(\rho)}{\partial \rho} (\rho - \rho_0)$$

$$\frac{\partial}{\partial \rho} (\mathbf{H}\mathbf{B}(\rho)\mathbf{H}^{\top} + \mathbf{R})^{-1} = - \underbrace{(\mathbf{H}\mathbf{B}\mathbf{H}^{\top} + \mathbf{R})^{-1}}_{\text{done}} \mathbf{H} \frac{\partial \mathbf{B}(\rho)}{\partial \rho} \mathbf{H}^{\top} \underbrace{(\mathbf{H}\mathbf{B}\mathbf{H}^{\top} + \mathbf{R})^{-1}}_{\text{done}}$$

The Taylor series can be easily estimated from stuff we already computed.

Prior sensitivity (Tentative preliminary results!)

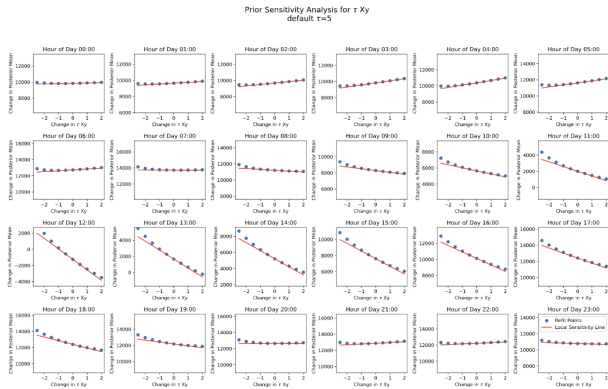


Figure 20: Prior sensitivity with respect to τ_{xy} for a single inversion

Problem: In inversion problems, the prior is informative, and the model does smoothing. How is your data being used? What patterns are possible to identify?

Example question: Can I detect emissions from road pixels?

Problem: In inversion problems, the prior is informative, and the model does smoothing. How is your data being used? What patterns are possible to identify?

Example question: Can I detect emissions from road pixels?

Idea: Let $\mathbf{r}$ be an N -vector encoding road pixels.

$$\mathbf{r}_n = \begin{cases} 1 & \text{If observation } n \text{ was a road pixel} \\ 0 & \text{otherwise} \end{cases}$$

Suppose the flux had been not $\mathbf{x}$, but $\mathbf{x}' = \mathbf{x} + \epsilon \mathbf{r}$. That is, $\mathbf{x}'$ is a little more “roady” than $\mathbf{x}$.

Suppose I used $\mathbf{y}' = \mathbf{y} + \epsilon \mathbf{H} \mathbf{r}$ to estimate $\hat{\mathbf{x}}'$. How much more “roady” is $\hat{\mathbf{x}}'$ than $\hat{\mathbf{x}}$?

$$\begin{aligned} \frac{\text{Inferred road pixel change}}{\text{Actual road pixel change}} &= \frac{\mathbf{r}^\top (\hat{\mathbf{x}}' - \hat{\mathbf{x}})}{\mathbf{r}^\top (\mathbf{x}' - \mathbf{x})} = \frac{\mathbf{r}^\top (\hat{\mathbf{x}}' - \hat{\mathbf{x}})}{\epsilon \mathbf{r}^\top \mathbf{r}} \\ &= \frac{\mathbf{r}^\top \overbrace{(\mathbf{H} \mathbf{B})^\top (\mathbf{H} \mathbf{B} \mathbf{H}^\top + \mathbf{R})^{-1} \mathbf{H} \mathbf{r}}^{\text{done}}}{\mathbf{r}^\top \mathbf{r}} \end{aligned}$$

This is just the “gain matrix” queried for a particular output (here, “roadiness”).

The data influence can be easily estimated from stuff we already computed.

Data influence (Tentative preliminary results!)

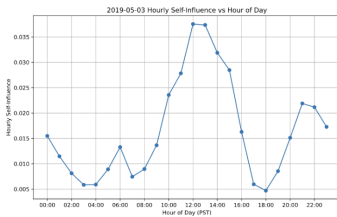


Figure 16: Hourly self-influence for a single inversion

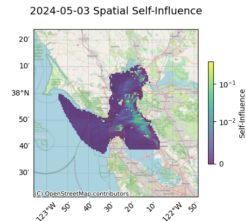


Figure 17: Spatial self-influence for a single inversion

Lots to do!